Step 1: Interface Task Selection & Empirical Problem Framing

1. **Target System Flow**
    a. The user flow that my project is about is the one that I will be selecting for this assignment. It specifically revolves around the offboarding process for an employee through the Office admin portal.
2. **Empirical Rationale**
    a. The thing that I am wanting to test is specifically the speed and ease at which admins are able to offboard an employee. Currently admins have to go to several different pages to achieve the result of a single employee leaving the company. I want to take this from several pages, to one more-informative page.
    b. My proposed redesign would gather everything that an admin would need to offboard an employee into one place so that they can see and manage every possibility at the same time. This would not only speed up the process by removing the need to navigate through multiple admin portals, but it would also reduce errors by admin's forgetting about one specific task for a user that might deviate from the standard offboarding procedure.

Step 2: Hypothesis Formulation

1. **Directional Hypotheses**
    a. Admins that use my wizard will offboard a user faster than by using the traditional method.
    b. Admins that use my wizard will make fewer mistakes by using my wizard.
2. **Null Hypotheses**
    a. There will be no discernable difference in offboarding an employee using my wizard vs the traditional method.
    b. There will be an equivalent amount of mistakes made when using my wizard when compared to the traditional method of offboarding.

Step 3: Variable Formulation & Operationalization

1. **Independent Variables**
    a. The only variable that I will be changing is the interface. My wizard vs the current Office admin portal.
2. **Dependent Variables**
    a. **Efficiency Metric:** Time to complete the task.
    b. **Effectiveness Metric**: Errors or missed steps. The admin could forget to give mailbox or OneDrive access.

   c. **Subjective Satisfaction Metric:** I want this to be an addition to the existing admin portal, so I think that a System Usability Scale would be best. Something like "would you use this tool for offboarding in the future?"
3. **Confounding Variables & Controls**
   a. Someone with prior Office admin portal experience could skew the results, so I would have to filter participants and group by like-experience levels.
   b. Since this is a computer task, the computer used for the test may influence the results. To mitigate this, I should ensure that all participants use similar devices.

Step 4: Experimental Design Architecture & Counterbalancing

1. **Subject Design Selection**
   a. For my study I would likely have to use within-subjects. This is because my sample pool is quite small. I don't have access to very many system administrators. In order to mitigate the learning curve, I would have to split the participants and have them perform the tasks in opposite orders.
2. **Participant Sampling Plan**
   a. The target userbase for my study is system administrators. General helpdesk employees may also be included, but they would need a general level of knowledge on the offboarding process. A sample size greater than 5 would be ideal, but I only have access to 5 individuals who meet the criteria.

Step 5: Ethical Protocol Safeguards

1. **Informed Consent & Ethics**
   a. This test is to measure the differences in speed and performance of the current Office admin portal and the new Offboarding Wizard. This test is to evaluate the interfaces, and not the users partaking in the study. Any difficulties in performing the tasks provided are to be factored in as a valued metric to further our evaluation of the current application. No information on individual participants will be recorded. The only two metrics that will be studied are time and efficiency.

Post-Mortem Reflection

1. Why is it critical in HCI empirical research to distinguish between statistical significance ($P < .05$) and practical significance (real-world clinical or operational impact) when evaluating interface performance metrics?
   a. I think it is important to distinguish between these two because they may not always be equivalent in importance. Using my task as an example:

offboarding an employee a whole minute faster might be nice, but forgetting to save their emails before the account gets deleted has a major impact on company performance. At the end of the day, the time it takes to do the task is important, but it matters more that you do it correctly. That isn't always the case though. If the task previously took a week to perform and I was able to cut it down to a minute, then I could probably work around the few times that I made a mistake in doing so.

2. If you chose a Within-Subjects design, what specific asymmetric transfer or carryover effects could occur between interface conditions, and how does your counterbalancing matrix mathematically isolate that bias?
    a. If someone were to start by using the wizard, they might get an idea of every process that you should do, and the order in which you should do them. That information could directly translate into knowledge on how to navigate the current Office admin portal. To fight this, I would need to separate the participants into two groups. One group that starts on the Office admin portal, and the other one that starts with the wizard. The purpose of my wizard is to make life for existing system administrators easier though, so I would likely put 3 people in the "Office admin center first" group and put the last 2 in the "wizard first" group.